INCH-POUND

MIL-O-82820(OS)
29 December 1989
SUPERSEDING
WS 20881
8 November 1983

MILITARY SPECIFICATION

OCTADECYL ISOCYANATE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 **Scope.** This specification defines one type of octadecyl isocyanate, referred to herein as ODI (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 **Government documents.**

2.1.1 **Specifications, standards and handbooks.** The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation
(see 6.2).

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6810

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STANDARDS

FEDERAL

FED-STD-313 Material Safety Data, Transportation Data and Disposal Data for Hazardous Materials Furnished to Government Activities

MILITARY

MIL-STD-129 Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Military Specifications and Standards, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 **Other Government documents, drawings, and publications.** The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

49 CFR 100-199 **Transportation**

(Application for copies of CFRs should be addressed to the Superintendent of Documents, U. S. Government Printing Office, Washington, D.C. 20402-0001.)

2.2 **Non-Government publications.** The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association Inc., Attn: Traffic Dept., 2200 Mill Road, Alexandria, VA 22314-5094.)

NATIONAL RAILROAD FREIGHT COMMITTEE

Uniform Freight Classification (UFC) 6000

(Application for copies should be addressed to the National Railroad Freight Committee, 222 South Riverside Plaza, Suite 1106, Chicago, IL 60606-5945.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 **First article.** When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.2 **Chemical and physical properties.** Chemical and physical properties of the ODI shall conform to the requirements specified in table I.

TABLE I. Chemical and physical properties.

Property	Limits	
	Minimum	Maximum
Amine equivalent, gram/equivalent	---	305
Hydrolyzable chloride (as hydrochloric acid), percent	---	0.01

3.3 **Infrared spectrum.** The infrared (IR) spectrum shall conform qualitatively to the spectrum shown on figure 1.

3.4 **Safety.**

3.4.1 **Toxic products and formulations.** The material shall have no adverse effect on the health of personnel when used for its intended purpose. Questions pertinent to this effect shall be referred by the contracting activity to the appropriate departmental medical service who will act as an advisor to the contracting agency (see 4.1.2).

3.4.2 **Material Safety Data Sheet (MSDS).** The contractor shall prepare and submit a MSDS in accordance with FED-STD-313 as specified in the contract (see 6.4).

3.5 **Shelf life.** The ODI shall meet the requirements of this specification for a period of 12 months from the date of quality conformance inspection (see 4.4) when packaged as specified in section 5 and stored at 40 degrees Fahrenheit (^{0}F) maximum.

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3.5.1 **Shelf life extension.** The shelf life expiration date of an individual lot may be extended for additional 12-month periods provided the ODI, upon retest, conforms to the amine equivalent requirement.

3.6 **Workmanship.** The workmanship shall be such that the ODI is free from contamination and defects.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 **Responsibility for compliance.** All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 **Toxicological product formulations.** The contractor shall have the toxicological product formulations and associated information available for review by the contracting activity to evaluate the safety of the material for the proposed use.

4.2 **Classification of inspections.** The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 **First article inspection.** First article inspection shall consist of the quality conformance inspections (see 4.4). Failure to meet all of the requirements specified herein shall be cause for rejection of the first article sample.

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4.4 **Quality conformance inspection.** Quality conformance inspection shall consist of the examination and tests specified in table II.

TABLE II. First article and quality conformance inspections.

Examination and tests	Requirement	Test method
Visual examination	3.6 and section 5	4.5.1
Amine equivalent test	3.2	4.5.2
Hydrolyzable chloride test	3.2	4.5.3
Infrared spectrum test	3.3.1	4.5.4

4.4.1 Sampling for inspection.

4.4.1.1 Lot. For inspection purposes, a lot shall consist of all the ODI subject to inspection at one time from one of the following:

a. A single batch (see 6.5.1)

b. A continuous process (see 6.5.2)

c. A uniform blend of batches or a uniform blend of portions from a continuous process.

4.4.1.2 Selection of sample containers. Sample containers of the ODI shall be selected at random from each lot in accordance with table III for the inspections specified in 4.5.1 through 4.5.4.

TABLE III. Sampling for inspection.

Number of containers in the lot	Number of sample containers to be selected for inspection
1 - 3	All
4 - 40	3
41 - 65	4
66 - 110	5

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4.4.1.3 **Sample material.** Sufficient material shall be selected at random from each sample container to perform all of the inspections specified in 4.5.1 through 4.5.4.

4.4.1.3.1 **Sample preparation.** The selected single samples from a lot shall be combined and blended to form three composite samples. An individual single sample shall not be split between composites except where a lot consists of fewer than three containers. A single determination of each quality characteristic listed in table I shall be performed on each of the three composite samples independently. When a lot consists of fewer than three containers, three samples shall be taken from each container. The sample portions obtained shall be grouped and blended to form three composite samples. The results of each test shall meet the requirements of table I. Failure of any composite sample to meet any requirement shall be cause for rejection of the lot.

4.5 **Methods of inspection.** The following inspection methods shall be used. Unless otherwise specified in the test or procedure description, all weights, volumes, temperatures, and times shall be measured to the nearest specified unit or decimal.

NOTE: Reagent grade chemicals shall be used for chemical reactions in the conduct of all tests defined herein. Solvents and indicators may be commercial nonreagent grade materials.

4.5.1 **Visual examination.** Each sample of material shall be visually examined for conformance to 3.6 and section 5.

4.5.2 **Amine equivalent test.** Amine equivalent shall be determined in accordance with the following:

a. Weigh 6.0 ± 0.5 grams (g) of sample to the nearest 0.001 g into a flask.

b. Dissolve the sample in 50.0 ml of dried toluene or 1,2,4 trichlorobenzene.

c. Pipette 20.0 milliliters (mL) DBA solution (prepared by diluting 338 ml di-n-butyl amine to one Liter with dried toluene) into the flask.

d. Mix thoroughly and allow the stoppered flask to stand at room temperature for 15 to 20 minutes.

e. Add 100 ± 20 ml of methanol and 3 to 10 drops of 0.1 percent bromophenol blue indicator, and titrate with standardized 1N hydrochloric acid (HCl) to the appearance of a yellow color that persists for at least 15 seconds.

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f. Run a minimum of three blank determinations omitting only the sample.

g. Calculate the amine equivalent as follows:

$$\text{Amine equivalent, grams/equivalent} = \frac{1000\ W}{(B - A)\ N}$$

Where: A = volume of HCl for sample titration, mL

B = volume of HCl for blank titration, mL

N = normality of HCl

W = sample weight, g

1000 = conversion from milliequivalents to equivalents

h. Record amine equivalent to the nearest 0.1 gram/equivalent.

4.5.3 **Hydrolyzable chloride test.** Hydrolyzable chloride shall be determined in accordance with the following:

a. Weigh 5.0 ± 0.1 g of sample to the nearest 0.001 g in a beaker.

b. Dissolve the sample in approximately 100 mL of toluene, and add approximately 100 mL of methanol.

c. Heat (while gently stirring) to a gentle boil, and boil for 15 to 20 minutes.

d. Cool to room temperature.

e. Add 20 drops of concentrated nitric acid and titrate potentiometrically with standardized 0.01N silver nitrate using an automatic titrator with a silver billet electrode and a potassium nitrate-filled calomel reference operated in accordance with the manufacturer's instructions.

f. Calculate the hydrolyzable chloride as follows:

$$\text{Percent hydrolyzable chloride (as \% hydrochloric acid)} = \frac{AB\ 3.646}{W}$$

Where: A = volume of standardized silver nitrate, mL

B = normality of standardized silver nitrate

W = sample weight, g

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3.646 = one-tenth of the molecular weight of hydrochloric acid

g. Record the hydrolyzable chloride to the nearest 0.001 percent.

4.5.4 **Infrared spectrum test.** Infrared spectrum shall be determined using an infrared spectrophotometer in accordance with the following:

a. Smear a sample of ODI on a demountable infrared sodium chloride window.

b. Adjust the smear thickness to obtain a spectrum with a background above 80 percent transmittance and the major peaks between 5 and 30 percent transmittance.

c. Scan the sample between 4000 and 700 reciprocal centimeters.

d. Compare the sample spectrum with figure 1, and record any deviation in the position and magnitude of the major peaks.

4.6 **Inspection of packaging.** The ODI containers and container markings shall be visually examined to verify compliance with section 5.

5. PACKAGING

5.1 **Packaging.** Unless otherwise specified in the contract or order (see 6.2), packaging shall be level C as specified herein.

5.1.1 **Level C.** The ODI shall be packaged to afford adequate protection against loss and contamination during shipment from the supply source to the first receiving activity and during storage under the shelf life period and conditions specified in 3.5. Containers in the same shipment shall be of the same size. The packaging shall conform to UFC 6000, National Motor Freight Classification, 49 CFR 171-178, or to other carrier rules and regulations as applicable to the mode of transportation.

5.2 **Marking.**

5.2.1 **Standard marking.** In addition to any special marking required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

5.2.2 **Special marking.** In addition to the marking requirements of 5.2.1, each container marking shall include the following information:

a. Title, number and date of this specification.

b. Manufacturer's name and address.

c. Material trade name and identification.

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d. Net weight of contents.

e. Lot number and date of quality conformance inspection (see 3.5).

f. Storage conditions.

g. Toxicity warning (if applicable).

h. Shelf life and inspection/test date for extension. In addition, the marking shall include the following:

"SHELF LIFE EXTENSION

THE SHELF LIFE OF AN INDIVIDUAL LOT MAY BE EXTENDED FOR AN ADDITIONAL 12 MONTH PERIOD PROVIDED THE ODI, UPON RETEST, CONFORMS TO THE AMINE EQUIVALENT REQUIREMENTS SPECIFIED IN MIL-O-82820."

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 **Intended use.** The ODI is intended for use as a bonding agent in the manufacture of propellant for the Standard Missile Mark 104 Dual Thrust Rocket Motor.

6.2 **Acquisition requirements.** Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

c. Whether first article is required (see 3.1 and 6.3).

d. Material Safety Data Sheet (see 3.4.2 and 6.4).

e. Packaging required if other than as specified in 5.1.1.

f. Special marking if other than as specified in 5.2.2.

6.3 **First article.** When first article inspection is required, the contracting officer should provide specific guidance to offerors on the material to be inspected. The material should be a first article sample and should consist of a quantity of the ODI sufficient to perform all of the inspections specified in 4.3. The first article sample should be packaged and

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marked as specified in section 5. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 **Material Safety Data Sheet (MSDS).** MSDS requirements are applicable to this specification and should be specified in the contract as required by the Federal Acquisition Regulation (FAR), Part 23, Subpart 23.3. Contracting officers will identify those activities requiring copies of completed MDSDs prepared in accordance with FED-STD-313. The pertinent Government mailing addresses for submission of data are listed in FED-STD-313.

6.5 **Definitions.**

6.5.1 **Batch.** A batch is that amount of ODI made as one unit in the manufacturing process.

6.5.2 **Continuous process.** A continuous process is an uninterrupted process of manufacture from which, in one unbroken flow, the ODI is accumulated for a lot.

6.6 **Subject term (key word) listing.**

Missile, Standard
Propellant
Rocket Motor, Dual Thrust, Mark 104

Preparing Activity:
NAVY-OS
(Project 6810-N055)

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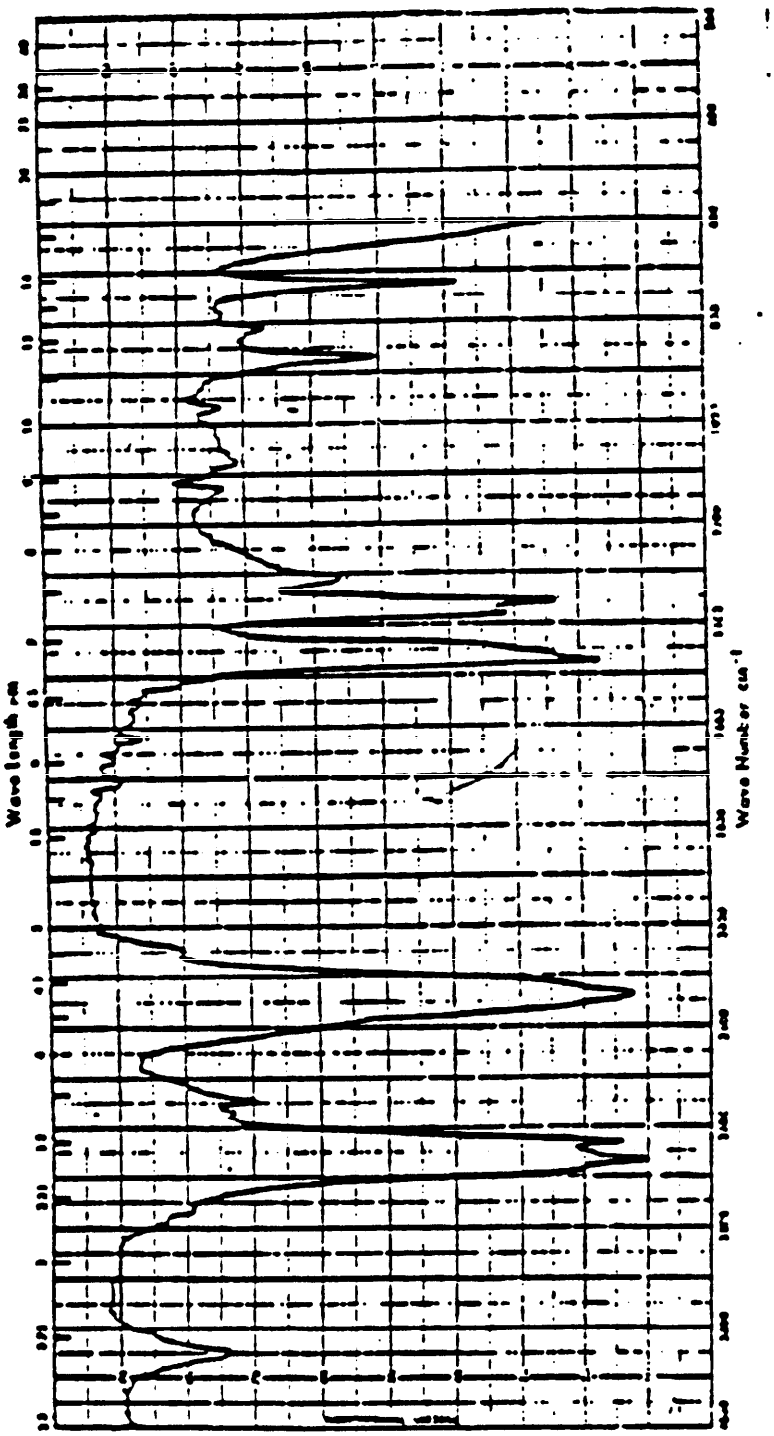


FIGURE 1. Infrared spectrum, octadecyl isocyanate.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-O-82820	2. DOCUMENT DATE (YYMMDD) 29 December 1989

3. DOCUMENT TITLE
OCTADECYL ISOCYANATE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)

7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY NAVAL ORDNANCE STATION (CODE 3730) INDIAN HEAD, MD 20640-5000

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON
c. ADDRESS (Include Zip Code)	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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